

Reference Sheet

Math

```
# import math library to use math functions
import math

result = math.ceil(6.1) # result == 7
result = math.floor(6.1) # result == 6
result = abs(-7) # result == 7
result = math.sqrt(9) # result == 3.0

# trigonometry
theta = math.pi # theta == pi
result = math.sin(theta) # result == 0.0
```

Arithmetic Operators

```
# add, subtract, multiply, divide, exponentiate
result = 20 + 25 # result == 45
result = 20 - 25 # result == -5
result = 20 / 25 # result == 0.8
result = 4 / 2 # result == 2.0
result = 20 * 25 # result == 500
result = 2 ** 4 # result == 16
result = "a" + "b" # result == "ab"

# integer division
result = 20 // 25 # result == 0
result = 4 // 2 # result == 2
result = 4.0 // 2 # result == 2.0
result = 4 // 2.0 # result == 2.0

# modulus - returns the remainder after division
result = 20 % 2 # result == 0
result = 25 % 2 # result == 1
result = 25 % 20 # result == 5
```

Comparisons and Logical Operators

```
# Equal to `==`, not equal to `!=`
result = 20 == 25 # result == False
result = 20 != 25 # result == True

# Less than `<`, greater than `>`
result = 20 > 25 # result == False
result = 20 < 25 # result == True
result = 20 < 20 # result == False

# less than or equal to `<=`, greater than or equal to `>=`
result = 20 <= 20 # result == True

# combining booleans using 'and'
result = 20 == 25 and 20 > 25 # result == False
result = 20 != 25 and 20 < 25 # result == True

# combining booleans using 'or'
result = 20 == 25 or 20 > 25 # result == False
result = 20 != 25 or 20 > 25 # result == True

# something is `in` a list, something is `not in` a list
result = 25 in [20, 25] # result == True
result = 25 in [20, 26] # result == False
result = 25 not in [20, 26] # result == True
```

Type Conversions

```
result = int(4.2) # result == 4
result = float("4.2") # result == 4.2
result = str(4.2) # result == "4.2"
```

Numpy

```
import numpy as np
# Create an array from a list
xs_arr = np.array([1,2,3])
# Create an array from another array
ys_arr = 3*xs_arr + 2 # result: np.array([5,8,11])
# Add arrays
z_arr = xs_arr + ys_arr # result: np.array([6,10,14])
```

Looping

```
# i goes from 0 .. 9
for i in range(10):
# i goes from 3 .. 9
for i in range(3, 10):
# i gets the value 3, then 5, then 7, then 9
for i in range(3, 10, 2):
# fruit gets the value "apple", then "banana", then "orange"
for fruit in ["apple", "banana", "orange"]:
```

```
# fruit gets the value "apple", then "banana", then "orange"
# index gets the value 0, then 1, then 2
for index, fruit in enumerate(["apple", "banana", "orange"]):

# zip allows you to go through multiple lists simultaneously
for a, b in zip(a_list, b_list):
for a, b, c in zip(a_list, b_list, c_list):

# break to exit a for loop early
# continue restarts the loop with the following element
for some_var in some_list:
    if some_condition:
        break
    if other_condition:
        continue #restarts the loop with the next element
```

Functions

```
# An example function definition:
def foo(a, b): # function 'foo' takes two parameters 'a' and 'b'
    return a + b # returns the result of adding 'a' and 'b'

# Using the example function:
result = foo(1, 2) # result gets the value 3
result = foo("1", "2") # result gets the value "12"
```

Lists

```
# Create lists:
my_list = [] # Empty
my_list = ["apple", "banana", "orange"] # Initial values

# Access a specific element in a list using an index:
result = my_list[0] # result == "apple"
my_list[0] = "pear" # my_list == ["pear", "banana", "orange"]

# Use negative index to get value in list in reverse order:
result = my_list[-1] # result == "orange"

# Get the length of a list:
result = len(my_list) # result == 3

# Add something to the end of a list:
my_list.append("lime")

# Get the maximum / minimum / sum of a list's items:
number_list = [1, 2, 3]
result = max(number_list) # result == 3
result = min(number_list) # result == 1
result = sum(number_list) # result == 6

# Remove something from the end of a list:
result = number_list.pop() # result == 3, number_list == [1, 2]

# And then remove something from a given index of a list:
result = number_list.pop(0) # result == 1, number_list == [2]

# Sort/reverse a list:
number_list = [3, 2, 1]
number_list.sort() # number_list == [1, 2, 3]
number_list.reverse() # number_list == [3, 2, 1]
```

CSV File

```
# A CSV file has the following structure on each line
# column1, column2
import csv
with open("example.csv", 'r') as file:
    reader = csv.reader(file)
    for line in reader:
        print(line) # each line is a list: ["column1", "column2"]
```